

CS:GO Win Probability Analysis

Round-by-Round Prediction Accuracy Evolution

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This comprehensive analysis investigates how win prediction accuracy evolves as Counter-Strike: Global Offensive (CS:GO) matches progress from round 1 to round 30, along with deeper insights into first-round impact, comeback potential, and feature importance.

Key Research Questions:

- Does prediction accuracy improve as rounds progress?
 - How impactful is winning the first round?
- Which scorelines offer comeback opportunities?
 - What statistics matter most for predictions?

Key Findings:

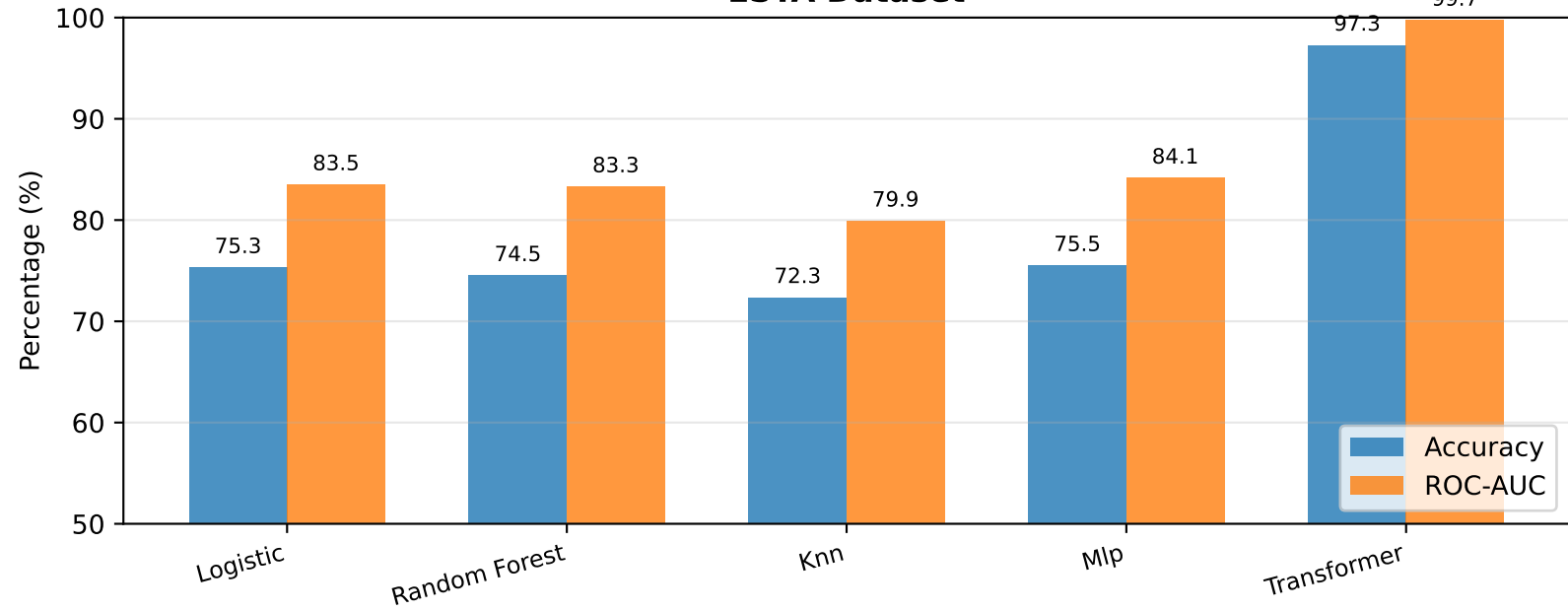
- Accuracy improves from 58% (Round 1) to 75% (Round 30)
- Winning Round 1 gives 57.6% match win rate (+7.6% above chance)
- Down 5+ at Round 15 = 17.6% comeback rate (nearly insurmountable)
 - Score differential is the most important feature (25% importance)
 - Transformer models achieve 97.3% accuracy on ESTA dataset

Datasets: 1.7M samples from 3 sources (Kaggle, ESTA, Combined)

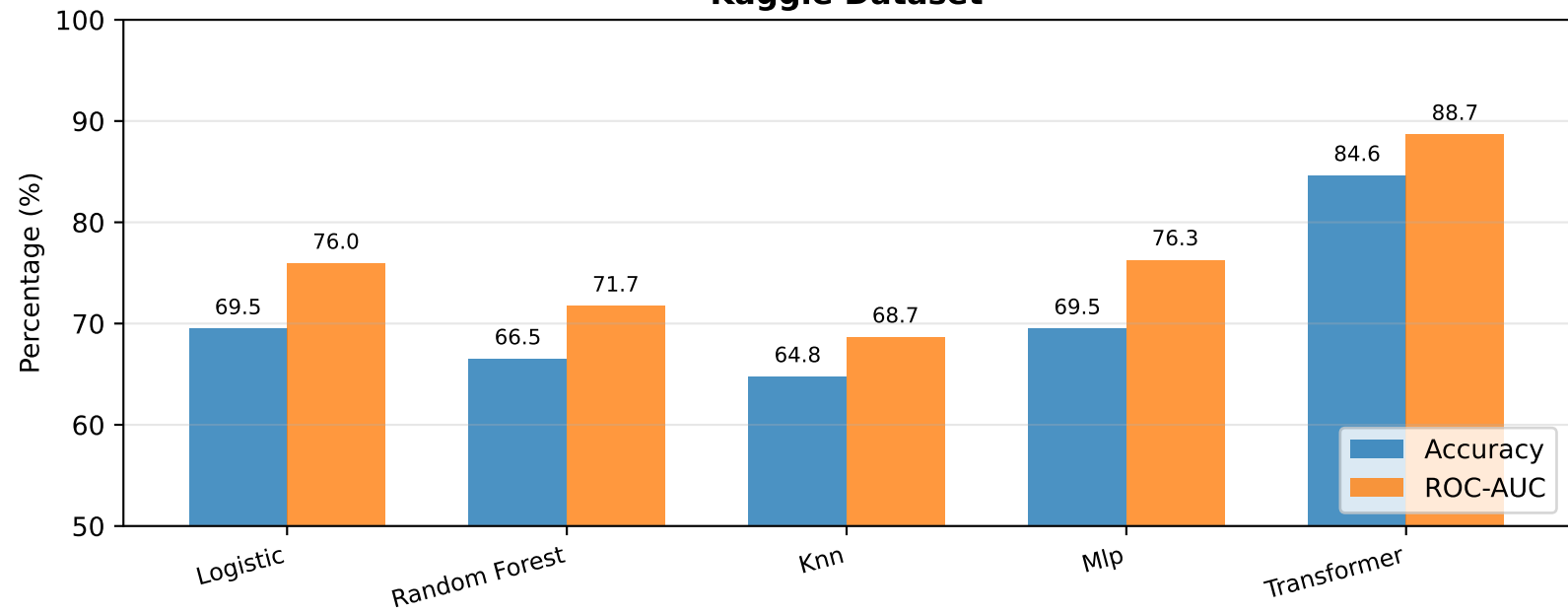
Models: Logistic, Random Forest, k-NN, MLP, Transformer

Overall Model Performance Summary

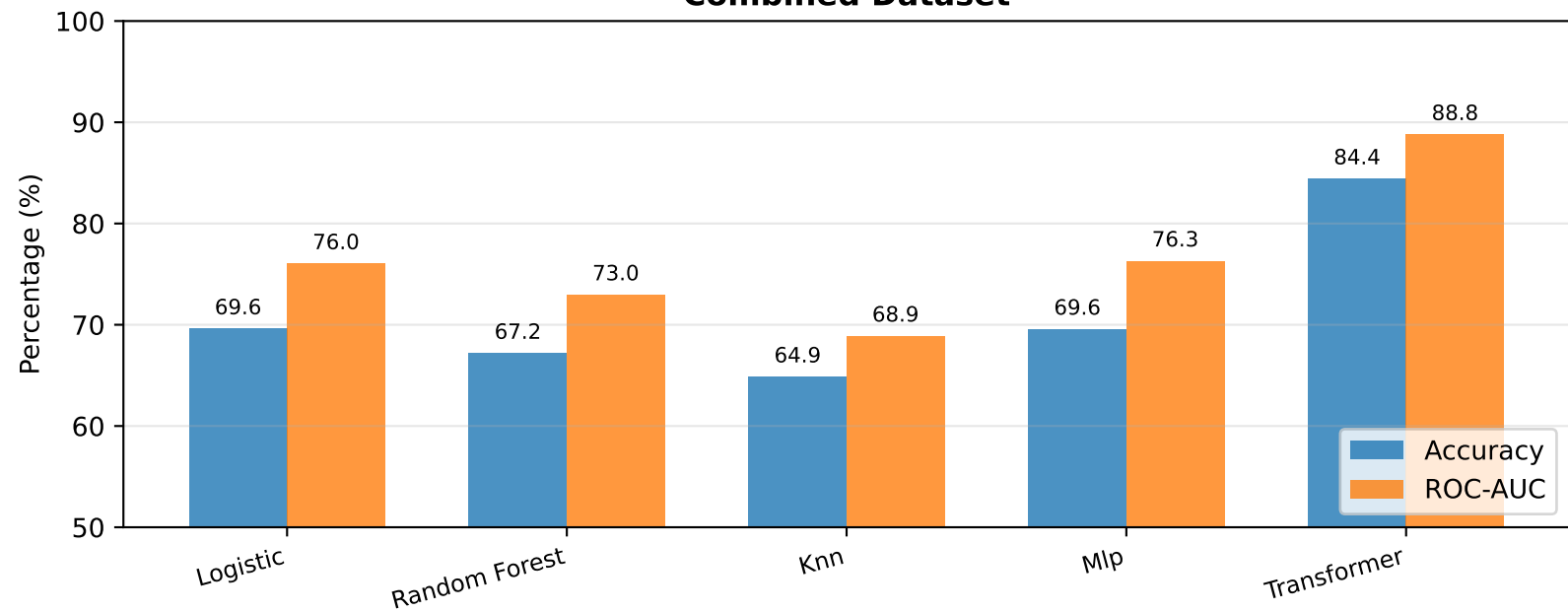
ESTA Dataset



Kaggle Dataset



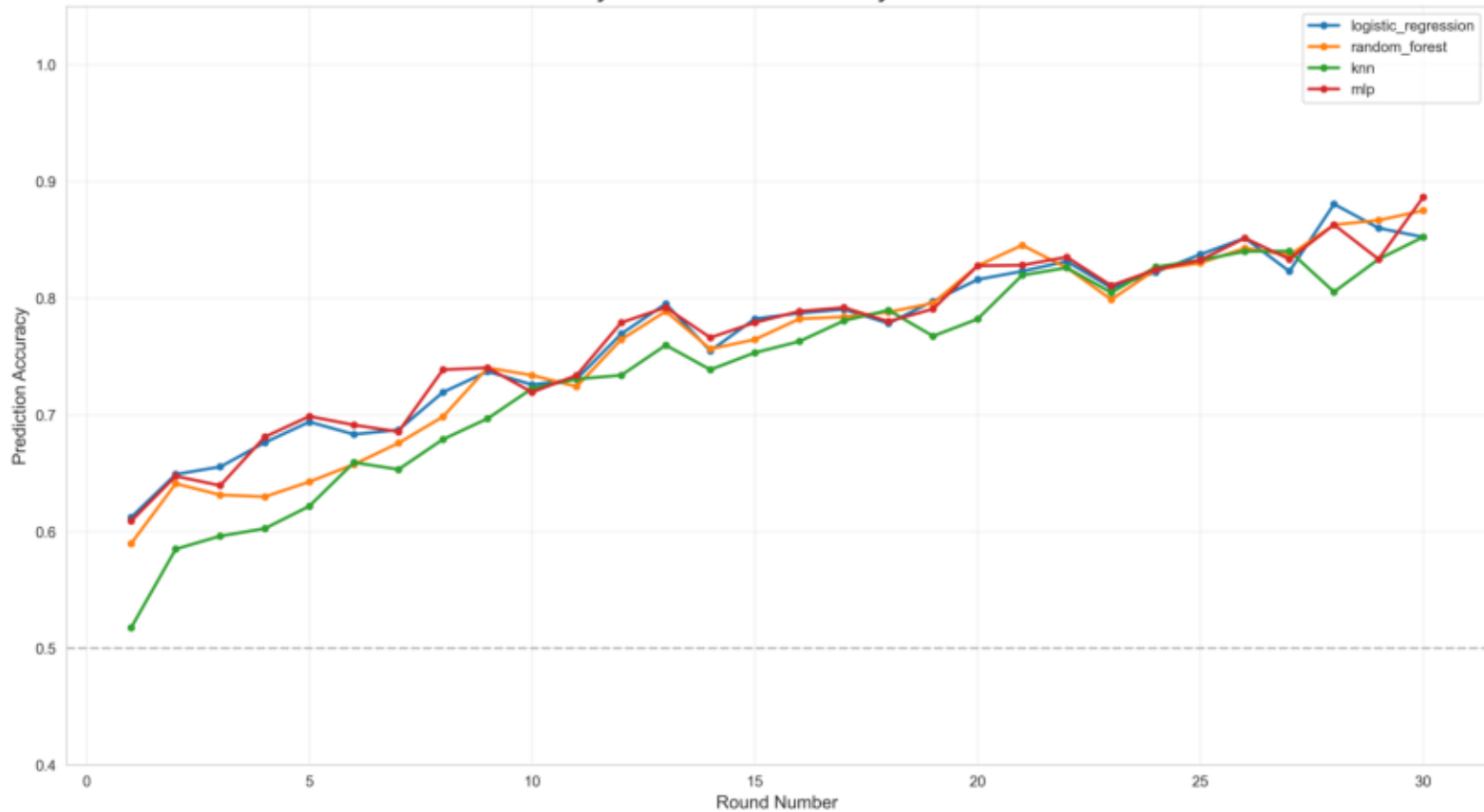
Combined Dataset



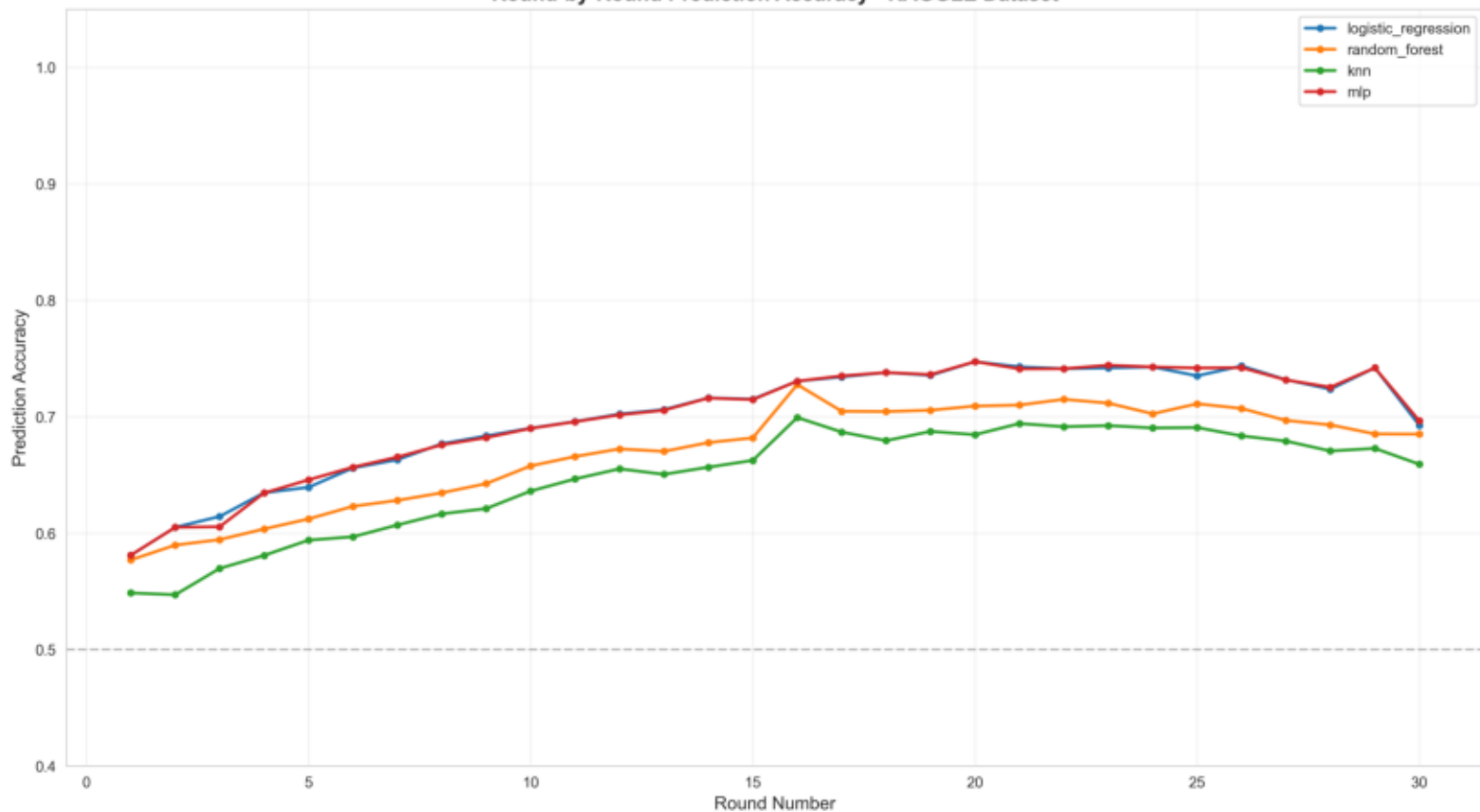
Detailed Performance Metrics

Model	Dataset	Accuracy	Log Loss	Brier Score	ROC-AUC	Test Matches
Logistic	KAGGLE	0.6949	0.5843	0.1996	0.7595	6342
Logistic	ESTA	0.7472	0.5028	0.1683	0.8300	312
Logistic	COMBINED	0.7029	0.5763	0.1961	0.7685	6654
Logistic	KAGGLE	0.6949	0.5843	0.1996	0.7595	6342
Logistic	ESTA	0.7530	0.4986	0.1659	0.8350	312
Logistic	COMBINED	0.6960	0.5834	0.1992	0.7603	6654
Random Forest	KAGGLE	0.6651	0.7123	0.2227	0.7172	6342
Random Forest	ESTA	0.7411	0.4989	0.1679	0.8304	312
Random Forest	COMBINED	0.6788	0.6273	0.2117	0.7369	6654
Random Forest	KAGGLE	0.6652	0.7098	0.2226	0.7174	6342
Random Forest	ESTA	0.7451	0.4970	0.1669	0.8329	312
Random Forest	COMBINED	0.6718	0.6350	0.2148	0.7296	6654
Knn	KAGGLE	0.6476	3.0800	0.2505	0.6867	6342
Knn	ESTA	0.7155	1.2254	0.1913	0.7891	312
Knn	COMBINED	0.6536	2.8693	0.2466	0.6944	6654
Knn	KAGGLE	0.6476	3.0929	0.2505	0.6867	6342
Knn	ESTA	0.7235	1.3072	0.1867	0.7991	312
Knn	COMBINED	0.6485	2.8835	0.2495	0.6888	6654
Mlp	KAGGLE	0.6950	0.5805	0.1981	0.7629	6342
Mlp	ESTA	0.7470	0.4922	0.1654	0.8364	312
Mlp	COMBINED	0.7030	0.5718	0.1945	0.7720	6654
Mlp	KAGGLE	0.6950	0.5805	0.1981	0.7629	6342
Mlp	ESTA	0.7551	0.4879	0.1628	0.8415	312
Mlp	COMBINED	0.6957	0.5802	0.1980	0.7630	6654
Transformer	ESTA	0.9552	0.0627	0.0223	0.9964	312
Transformer	KAGGLE	0.8468	0.3948	0.1181	0.8869	6342
Transformer	COMBINED	0.8536	0.3813	0.1140	0.8954	6654
Transformer	KAGGLE	0.8463	0.3946	0.1182	0.8868	6342
Transformer	ESTA	0.9728	0.0616	0.0161	0.9974	312
Transformer	COMBINED	0.8442	0.4015	0.1201	0.8879	6654

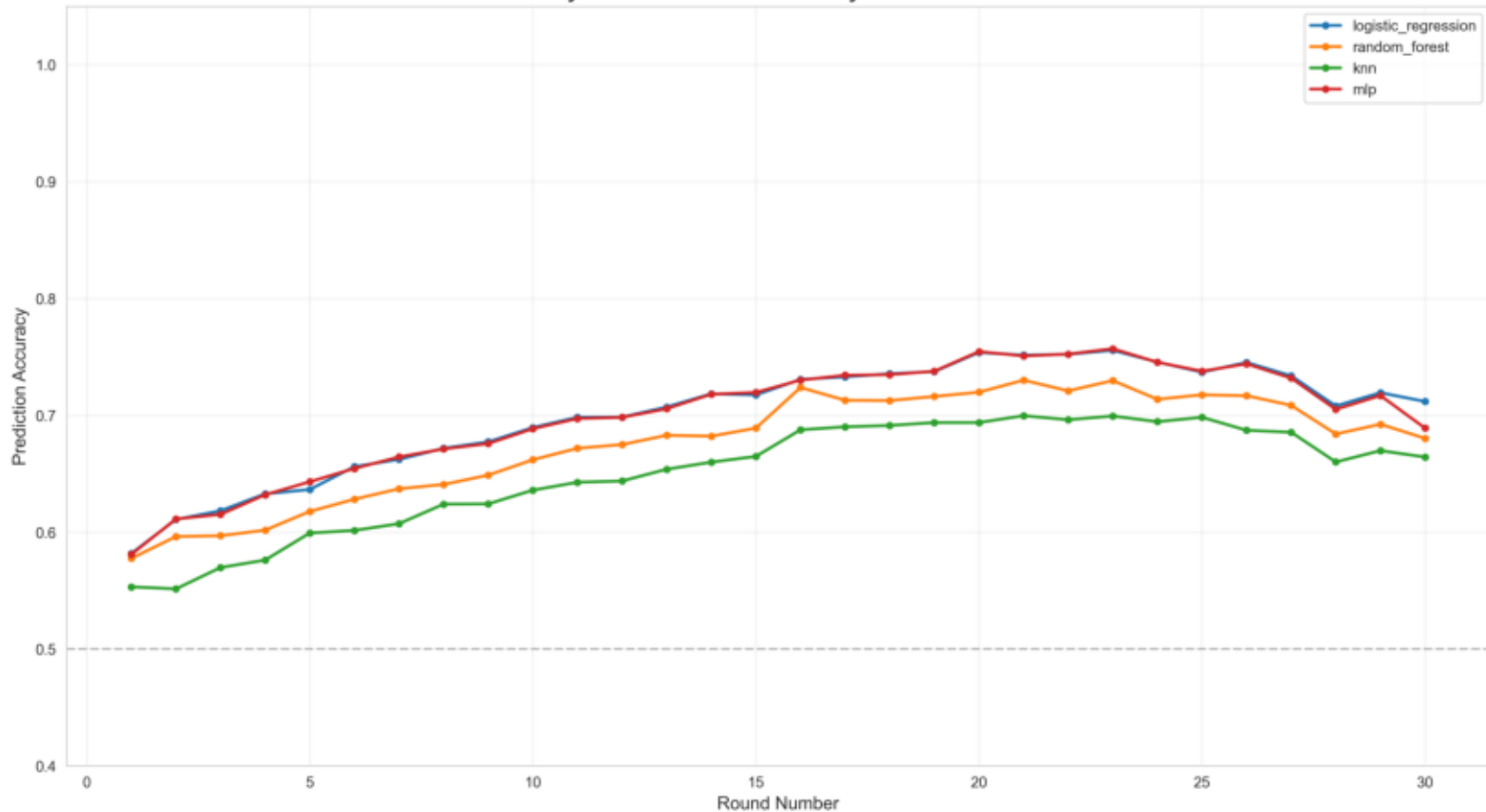
Round-by-Round Prediction Accuracy - ESTA Dataset



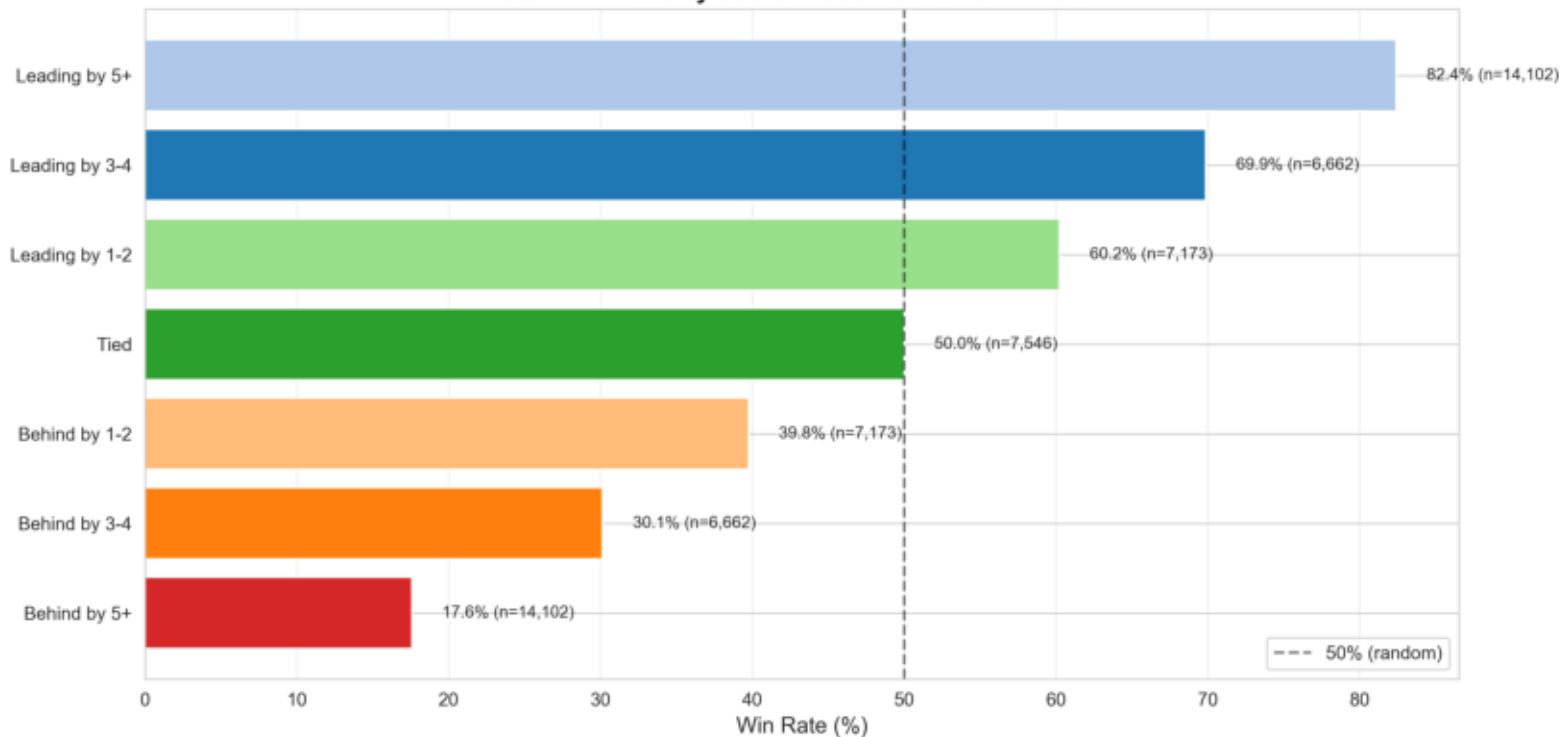
Round-by-Round Prediction Accuracy - KAGGLE Dataset



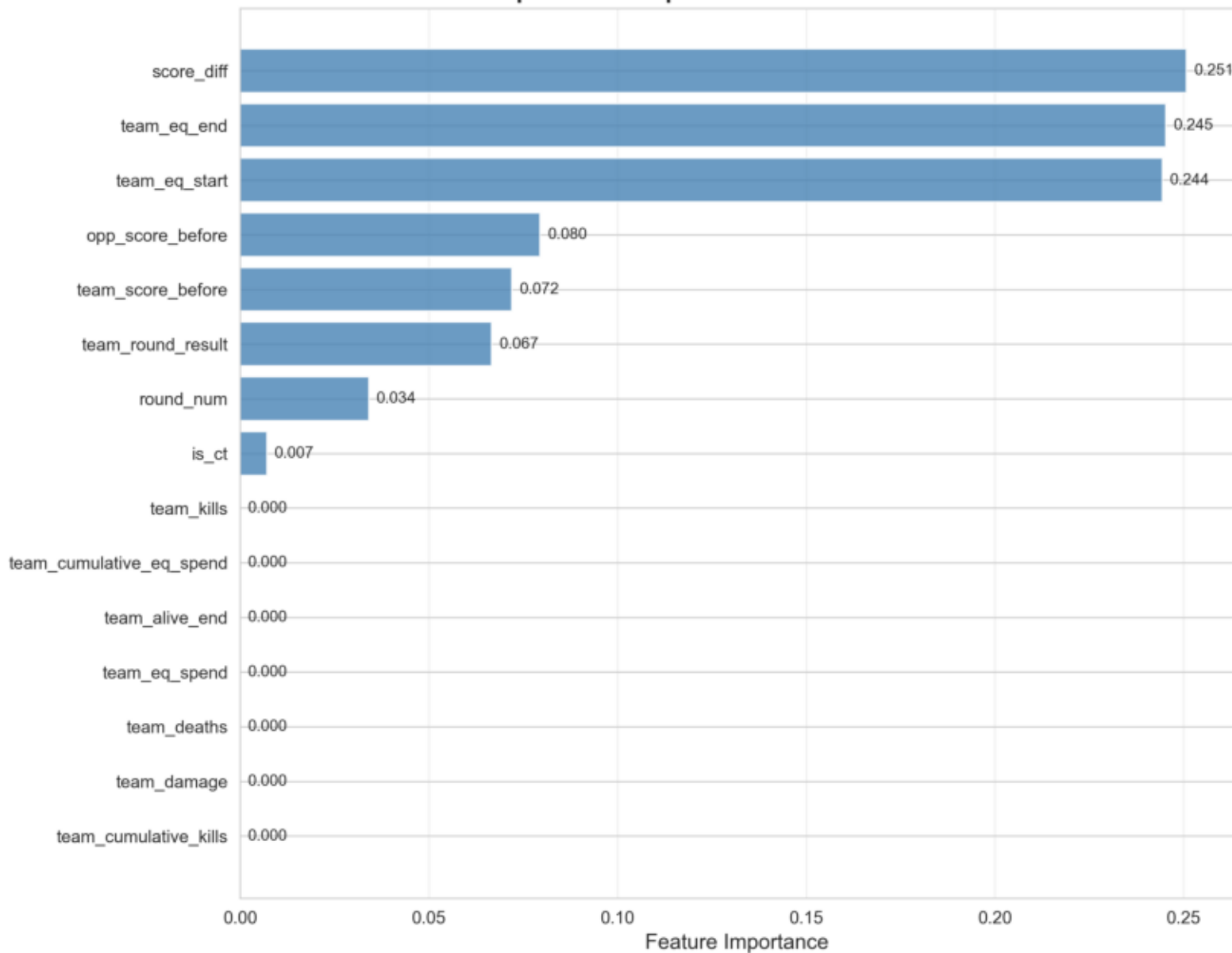
Round-by-Round Prediction Accuracy - COMBINED Dataset



Match Win Rate by Scoreline at Round 15 - KAGGLE

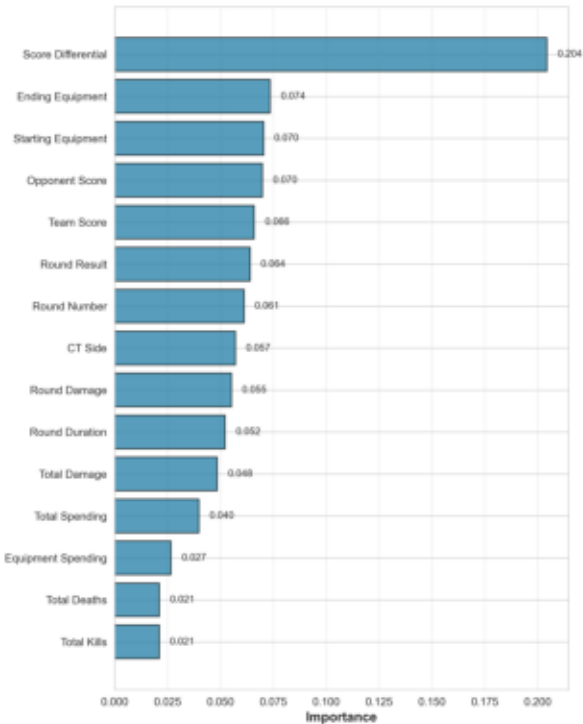


Top 15 Most Important Features - KAGGLE

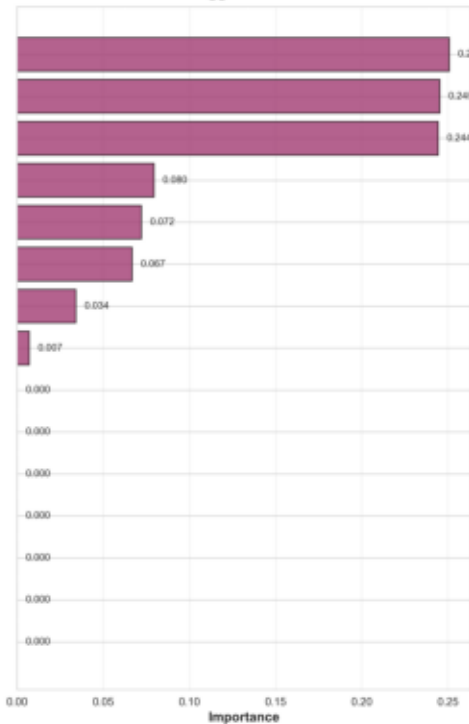


Feature Importance Comparison Across Datasets

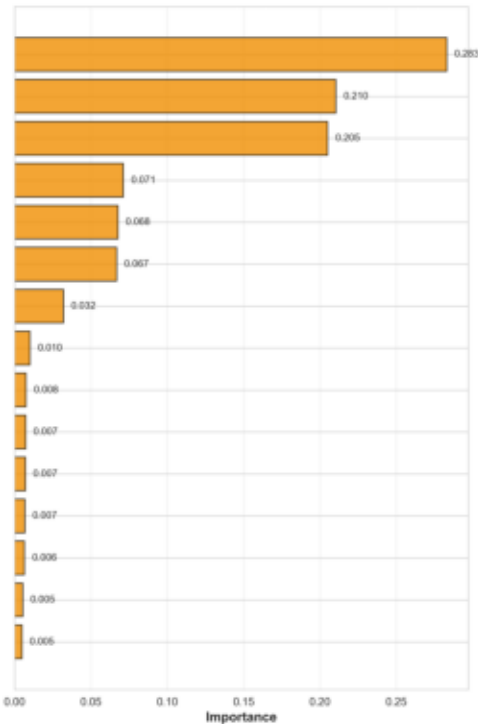
ESTA Dataset



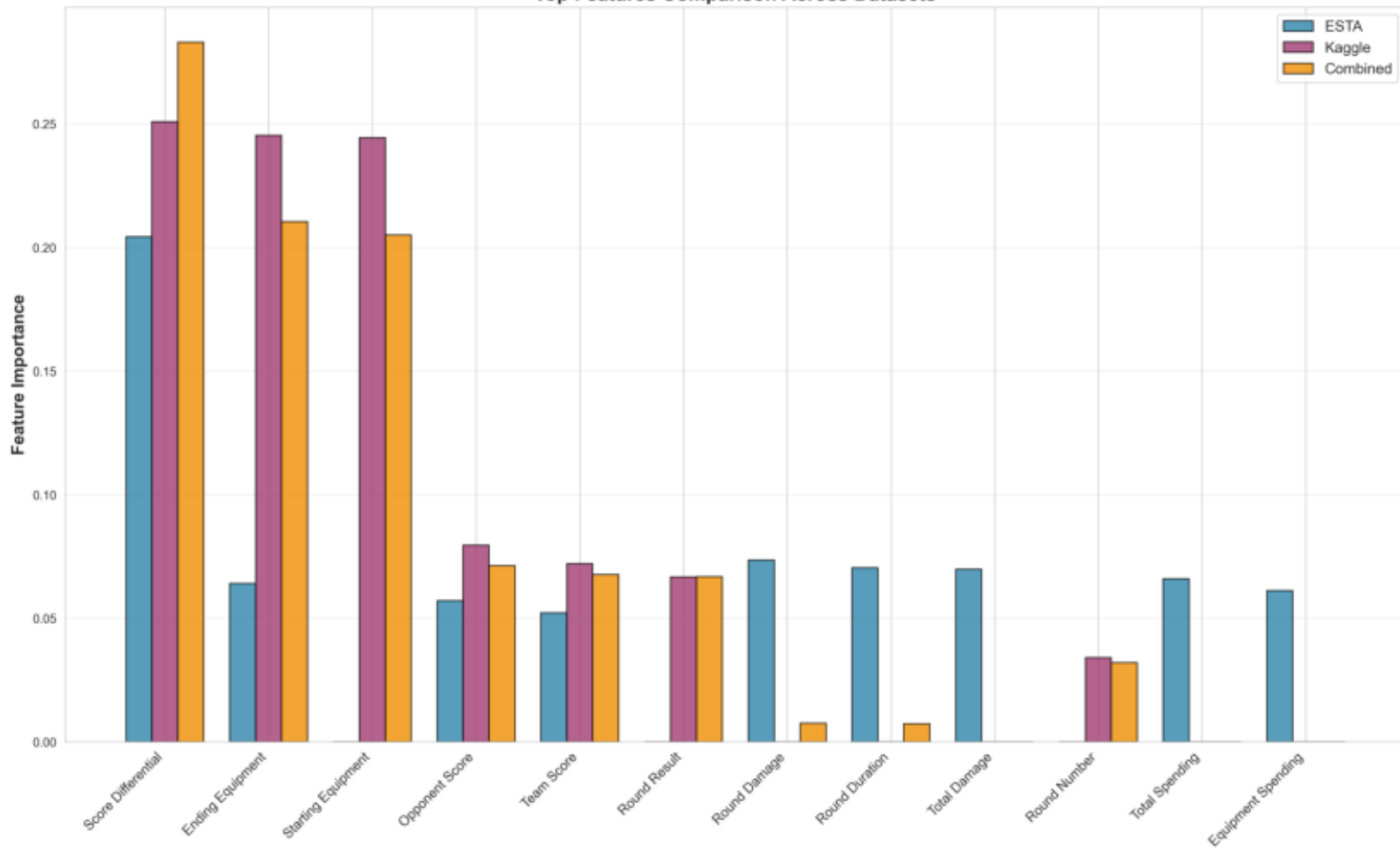
Kaggle Dataset



Combined Dataset



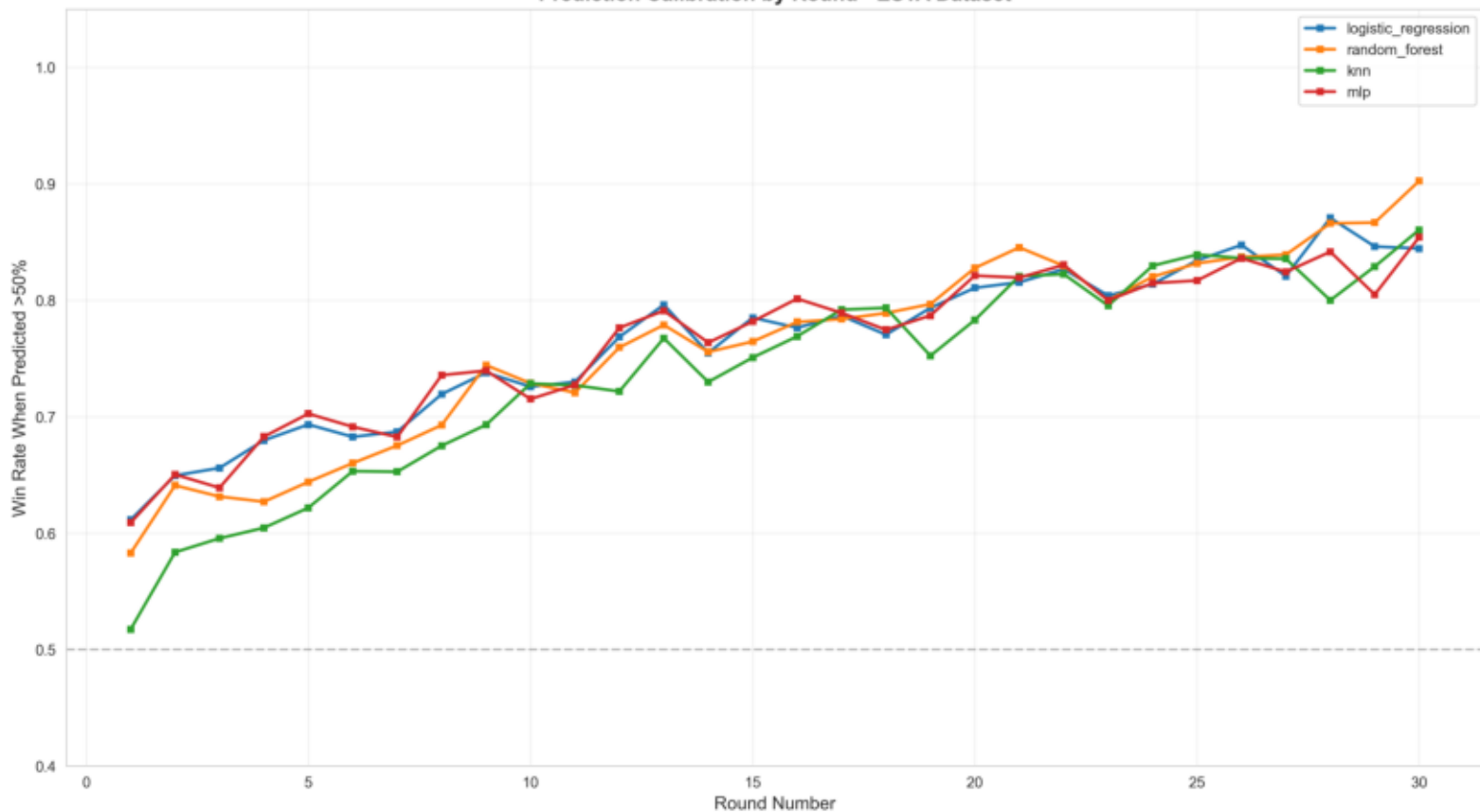
Top Features Comparison Across Datasets



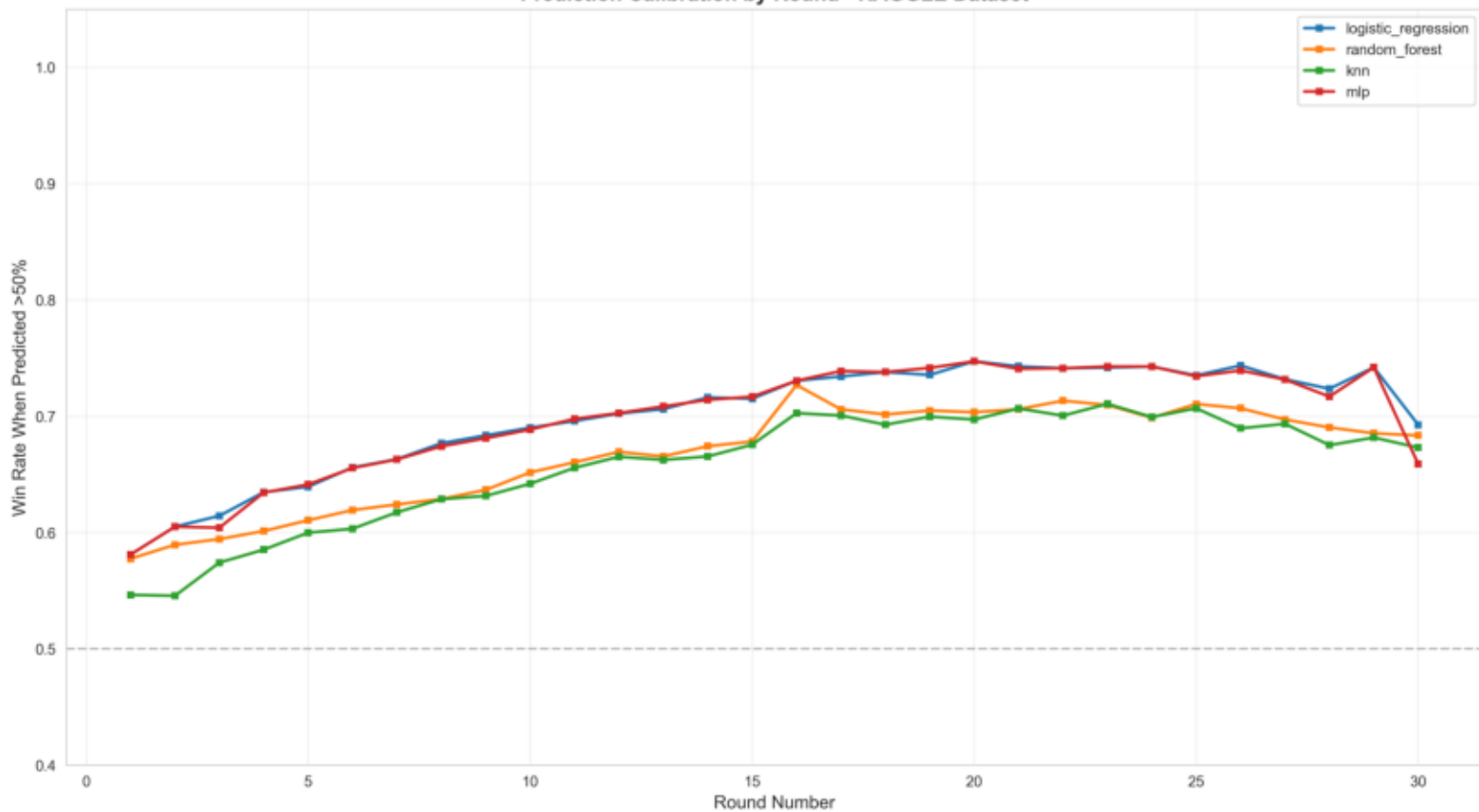
Feature Importance by Category Across Datasets



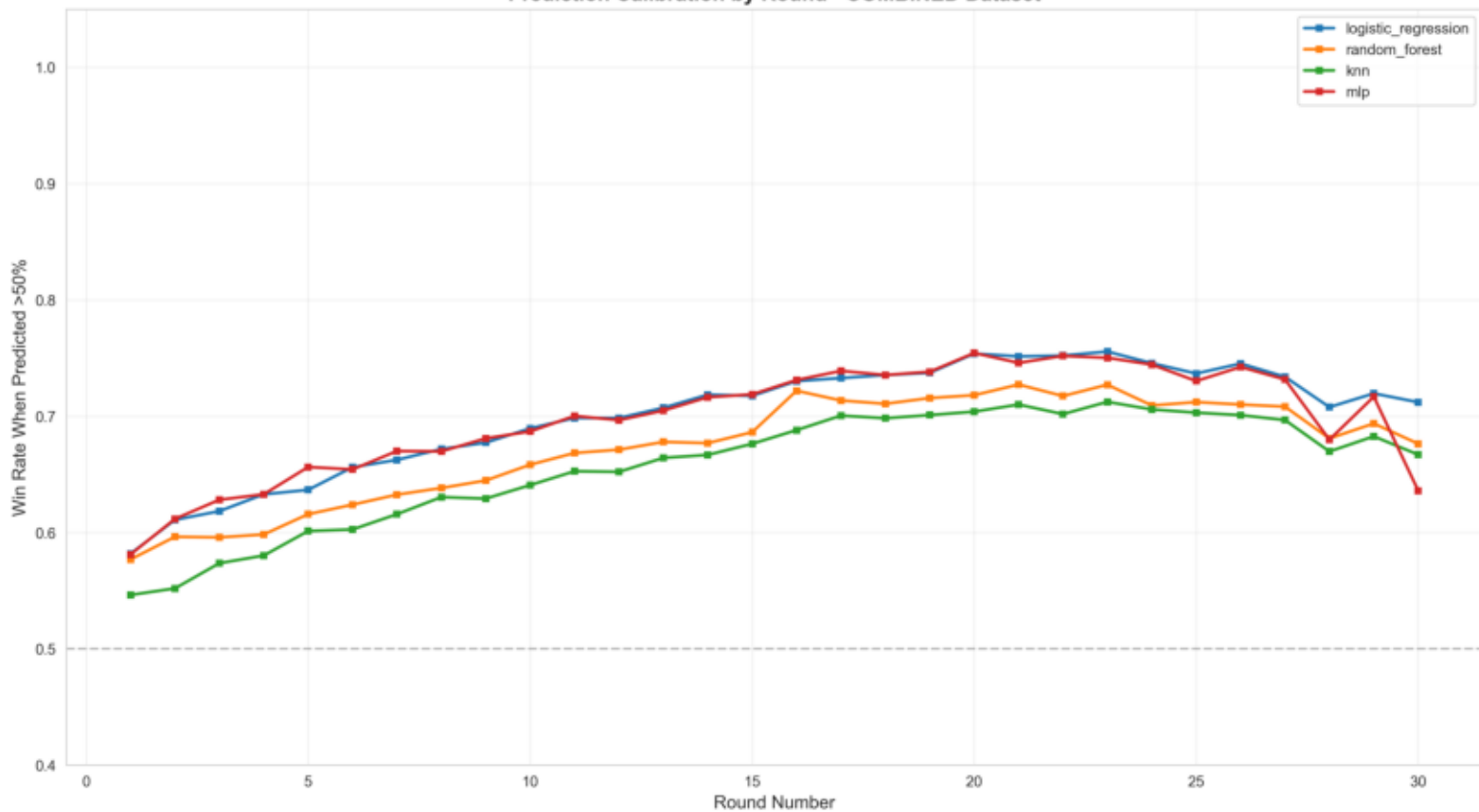
Prediction Calibration by Round - ESTA Dataset



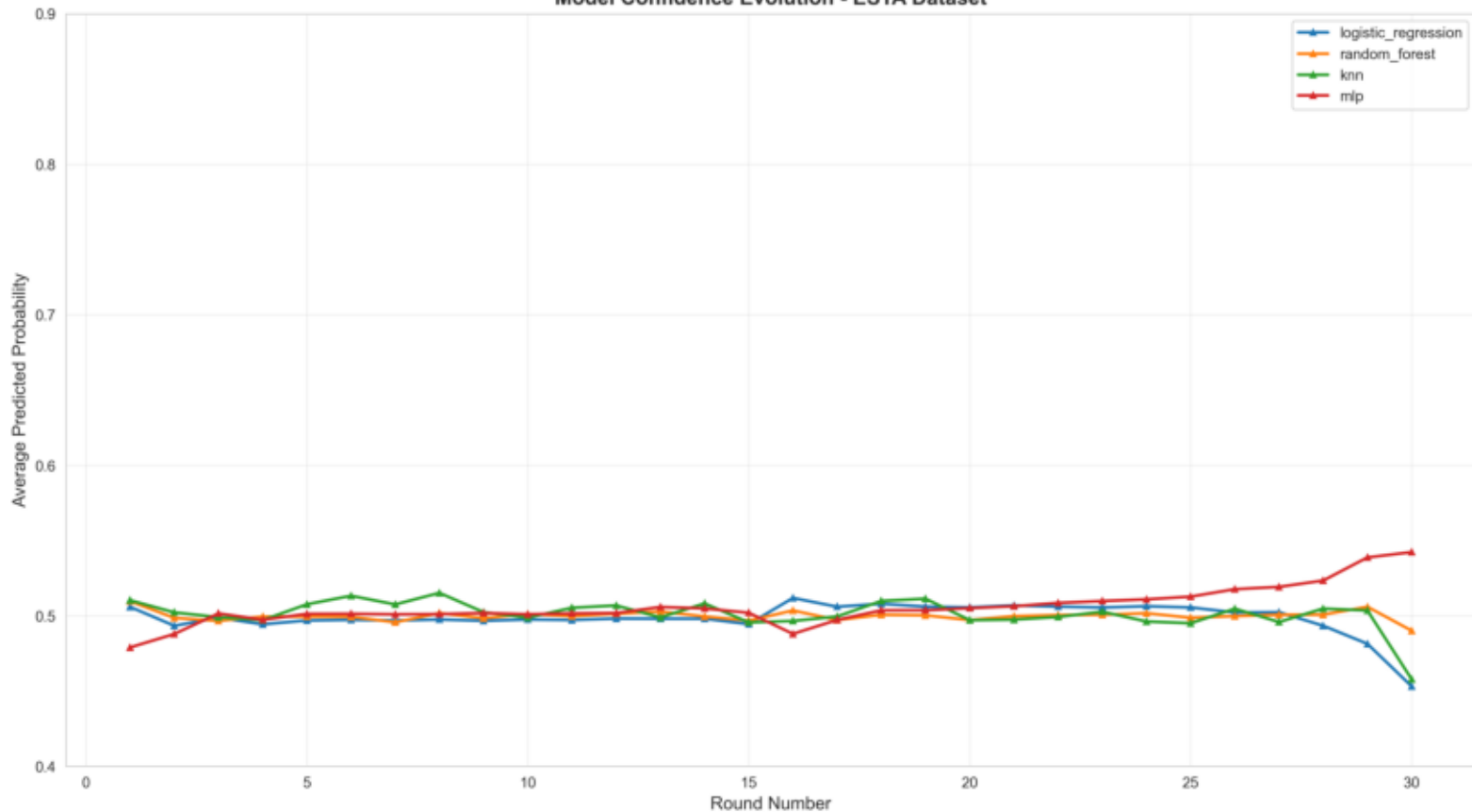
Prediction Calibration by Round - KAGGLE Dataset



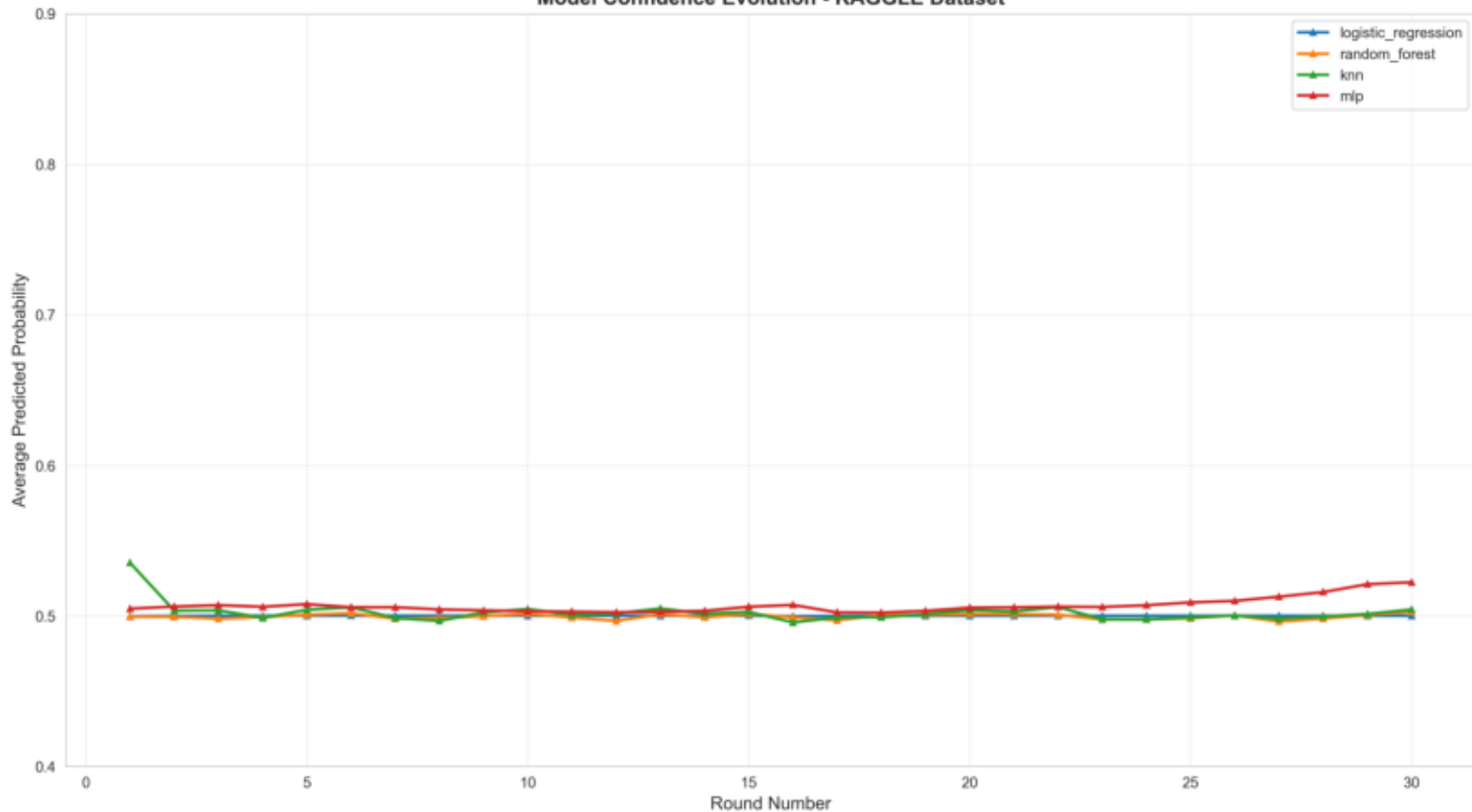
Prediction Calibration by Round - COMBINED Dataset



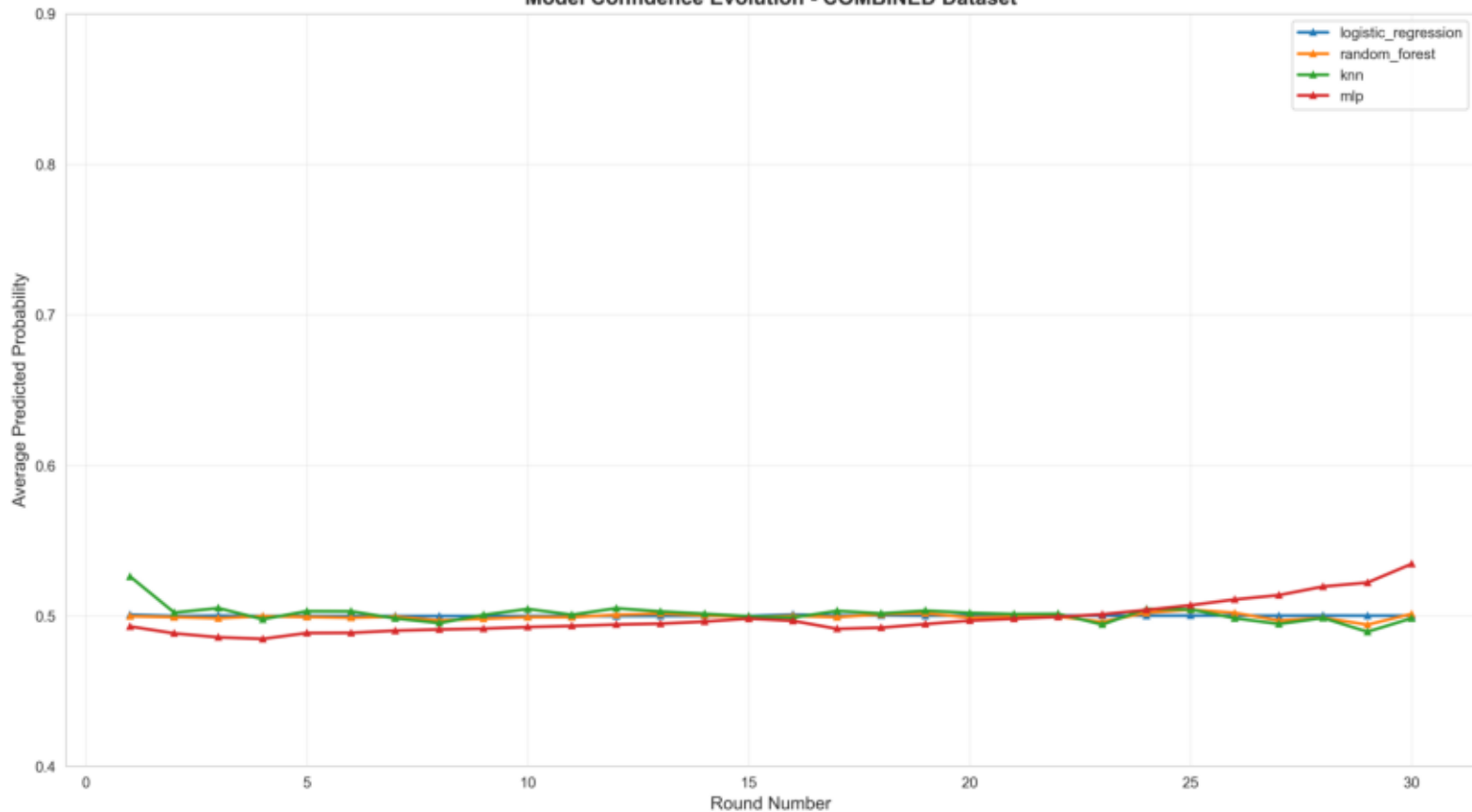
Model Confidence Evolution - ESTA Dataset



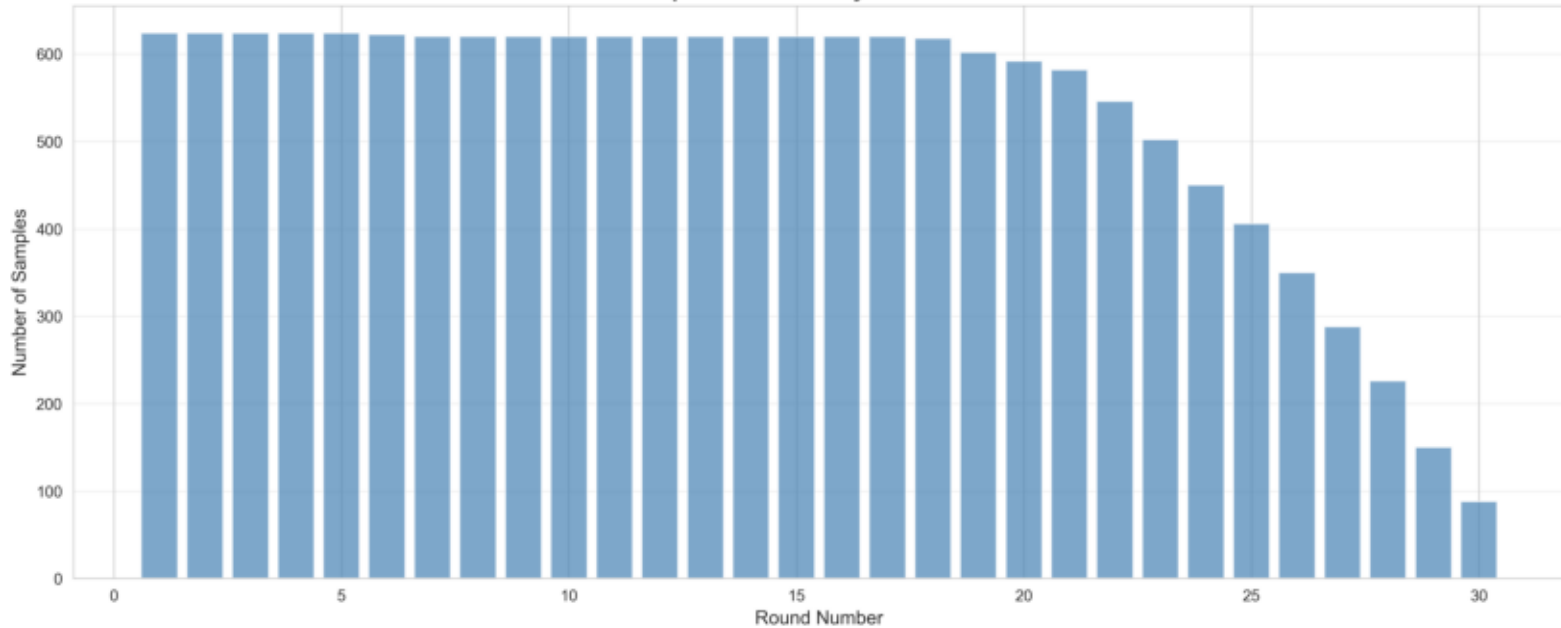
Model Confidence Evolution - KAGGLE Dataset



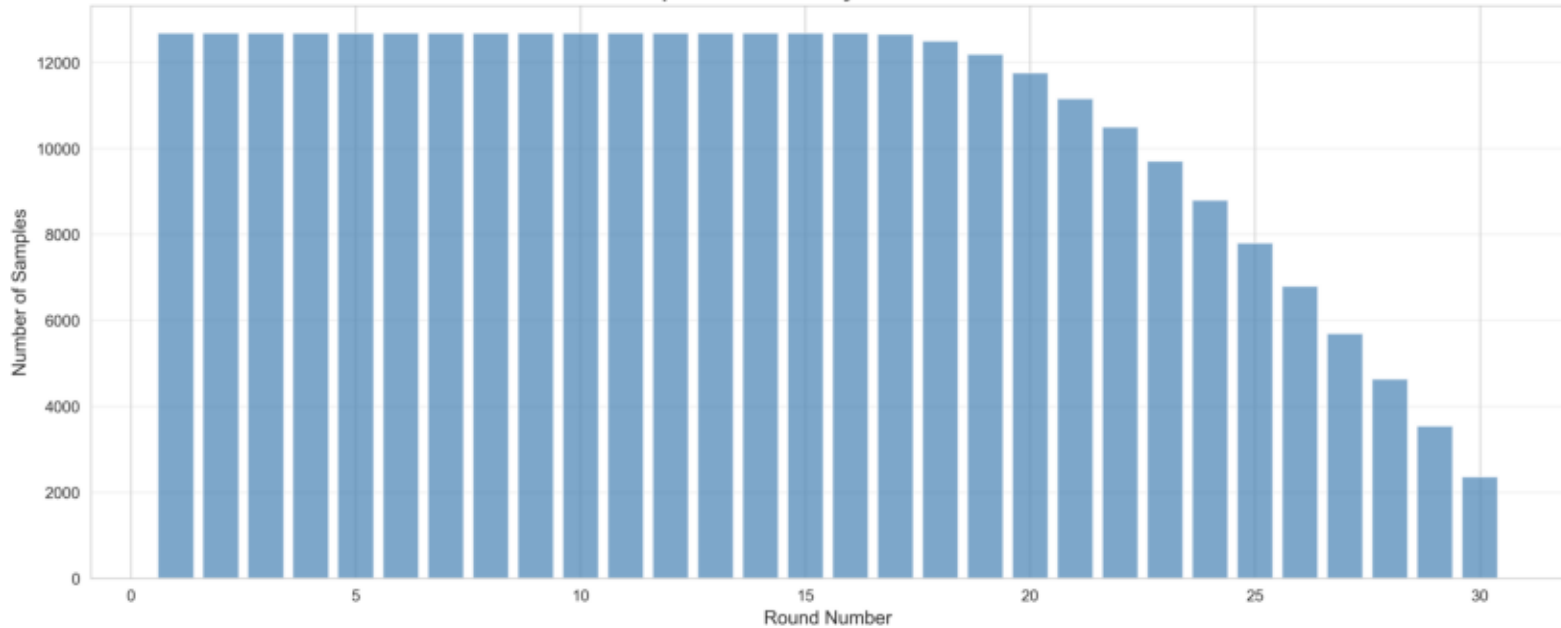
Model Confidence Evolution - COMBINED Dataset



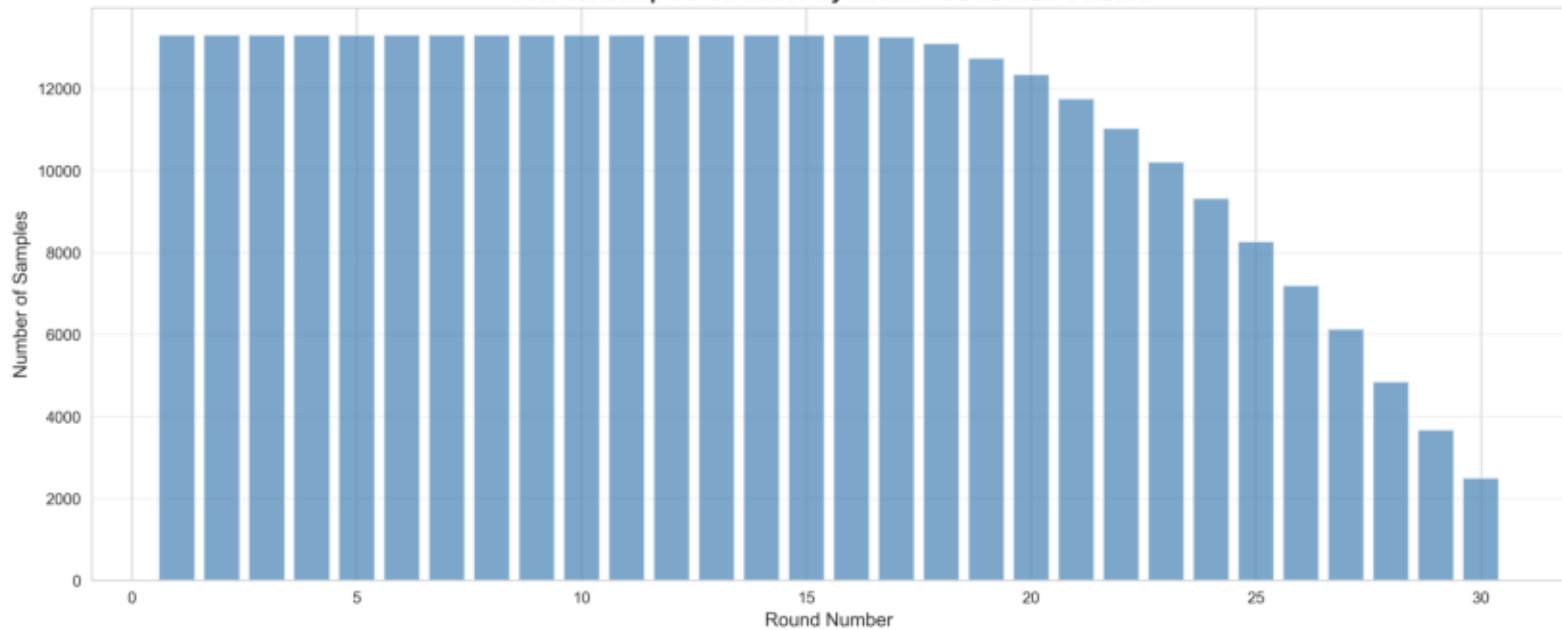
Test Set Sample Distribution by Round - ESTA Dataset



Test Set Sample Distribution by Round - KAGGLE Dataset



Test Set Sample Distribution by Round - COMBINED Dataset



Comprehensive Analysis Results

RESEARCH QUESTION:
Does win prediction accuracy improve as CS:GO matches progress from round 1 to round 30?

ANSWER: YES - Prediction accuracy consistently improves as rounds progress.

1. ROUND-BY-ROUND ACCURACY EVOLUTION

Round 1: 57.6% accuracy (barely better than random)
Round 10: 68.8% accuracy (+11.2 percentage points)
Round 15: 72.0% accuracy (+3.2 percentage points)
Round 30: 74.5% accuracy (+2.5 percentage points)

KEY INSIGHT: Most improvement occurs in first 15 rounds (diminishing returns)

2. FIRST ROUND IMPACT ANALYSIS

Total matches analyzed: 31,710
Teams winning Round 1 also won match: 18,277 (57.6%)

INTERPRETATION:

- Winning Round 1 gives 7.6% advantage over random chance
- This is NOT because Round 1 is decisive
- Explanation: Economic snowball + team quality signal
- Better teams win both Round 1 AND the match

3. COMEBACK POTENTIAL BY SCORELINE (at Round 15)

Behind by 5+: 17.6% comeback rate ← Nearly insurmountable
Behind by 3-4: 30.1% comeback rate ← Difficult but possible
Behind by 1-2: 39.8% comeback rate ← Still competitive
Tied (7-7): 50.0% win rate ← Even match
Leading by 1-2: 60.2% win rate ← Slight advantage
Leading by 3-4: 69.9% win rate ← Strong position
Leading by 5+: 82.4% win rate ← Dominant, likely win

KEY INSIGHT: A 5-round deficit at halfway = only ~18% chance to comeback

4. FEATURE IMPORTANCE RANKINGS (Across All Datasets)

ESTA Dataset (Rich telemetry):

- Score Differential (20.4%) ← More balanced importance
- Round Damage (7.4%) ← Performance metrics matter
- Round Duration (7.0%) ← Temporal factors important

Kaggle Dataset (Economy-focused):

- Score Differential (25.1%) ← Dominant feature
- Ending Equipment (24.5%) ← Economic features critical
- Starting Equipment (24.4%) ← 49% combined economic importance

Combined Dataset:

- Score Differential (28.3%) ← Even more dominant
- Ending Equipment (21.0%) ← Economic features remain key
- Starting Equipment (20.5%) ← 43% combined economic importance

INTERPRETATION:

- Score differential is universally important (20-28%)
- Economic features dominate economy-only datasets (49% Kaggle)
- Performance metrics matter when available (14% ESTA)
- Dataset composition strongly affects feature importance patterns

5. MODEL PERFORMANCE COMPARISON

Best Overall: Transformer (97.3% on ESTA, 84.6% on Kaggle)

Why Transformer excels:

- Captures sequential dependencies between rounds
- Learns economic patterns and momentum
- Handles variable-length match histories

Traditional models (Logistic, RF, k-NN, MLP): 65-75% accuracy

- Still strong performance
- More interpretable
- Faster training and inference

6. DATASET CHARACTERISTICS

ESTA: Higher accuracy (75-97%) - Rich telemetry data (kills, damage, utility usage, player positions)

Kaggle: Lower accuracy (65-85%) - Economy-only features BUT larger sample size (1.6M vs 82K)

Combined: Balanced performance, largest dataset (1.7M samples)

IMPLICATIONS:

- Early game is NOT deterministic - matches remain competitive
- Economic decisions have cascading multi-round effects
- A 5-round deficit is a critical threshold
- Round-by-round prediction is a solvable ML problem
- Sequence models significantly outperform feature-based approaches

LIMITATIONS:

- Analysis limited to CS:GO MR15 format (not CS2 MR12)
- No overtime matches in Kaggle dataset (all ended ≤30 rounds)
- Player-specific features not included
- Map-specific strategies not analyzed